Ethics Packs:
Knowledge Graph Injection for Moral Reasoning in Language Models

A Pilot Study with a Measurement Confound

Thomas Edrington
Liberation Labs / Transparent Humboldt Coalition

Version 1.1, October 2026. Experiments run 30 May 2026.

AI Disclosure: AI contributors to this work include Nexus, Lyra and CC, implemented on the Claude architecture (Anthropic). Their contributions are detailed in the Contributions section below. Version 1.1's second reader and style gate were also Claude instances. T. Edrington accepts accountability as corresponding human author.

Abstract

Empty responses, not the intervention, set the sign of our first result. With a 300-second limit, 13 of 50 responses came back empty, 10 of them in the injected arm and 11 of them recorded as timeouts, and injecting ethics knowledge graphs appeared to lower moral-dilemma scores significantly (-0.137, paired p=0.035); on the 15 pairs where both responses completed, the change was $+0.020$, while dropping only the pairs with a recorded timeout leaves -0.051. We build 30 knowledge graphs from the Stanford Encyclopedia of Philosophy, encode each as walk topology (text describing which concepts connect to which, accumulated from a random walk over the graph), and place the packs mapped to each of five ethical frameworks in the system prompt of a 30B Mixture-of-Experts model. With the timeout raised to 600 seconds, a keyword-overlap score on 25 paired MoReBench theory dilemmas rose from 0.501 to 0.539 ($+0.038$; paired $t(24)$=1.34, p=0.19, d_z=0.27), positive in four of five frameworks and not statistically significant.[1]

[1] The scorer counts words from rubric criterion titles, so it cannot separate reasoning from vocabulary priming, and the design has no mismatched-pack arm, so it does not test whether matching packs to frameworks matters (Sections 4.2 and 4.4).

Correction note (version 1.1, October 2026). Version 1.0 (drafts in the lab's published-research repository on GitHub from 30 May to 5 June and again since 22 July 2026, revised on 24 and 28 July; its academic edition on Zenodo since September 2026, DOI 10.5281/zenodo.21754904, all versions 10.5281/zenodo.21754903) said that theory-matched packs "improve moral reasoning scores", that "theory matching matters", and that the methodology was "validated". None of these is supported. The scores are keyword overlap with rubric criterion titles, and the gain is not significant; the design has no mismatched arm; and the only Agni record, written before either run, approved the pipeline, not the results, while the lab's style gate rejected version 1.0 on 4 September 2026. On 5 June a commit from Thomas Edrington's account had moved the draft out of the publishing queue with a BLOCKER status (944f98e), and the lab's research status table, in its 28 June snapshot, gives the blocker as "n=5, keyword-matching confound"; on 22 July a commit from Nexus's account, made by a Claude Opus 4.6 session, restored its text with the commit message "All cleared Agni gate with fixes applied", and no record of that clearance has been found. Version 1.0 also said it reported both runs but gave one number from the first; did not report a silently empty injected response that produces the whole of one framework's negative result; described the scorer and the injection prompt inaccurately; overstated the benchmark's criteria count; called the SEP library "164 ethics-related categories" (it has 30) and its 36,668 words "tokens"; labelled the paired effect size Cohen's d (it is d_z); called Aristotelian's v4 change "the true effect" and Scanlonian's -0.033 "within noise"; said ethical reasoning "can be enhanced at inference time" and that "the graph teaches structure"; and gave wrong metadata for three cited papers: invented author lists for MoReBench and for Sun et al., whose title was also invented, and for the lab's own graph-topology paper a title it never had. This version corrects all of these. Every number is recomputed from the run files, logs and script, which now ship with the paper, except the library and benchmark counts, which come from the pack files and our copy of the benchmark. The defects were found by the lab's style gate (2026-09-04, and again in October 2026), a scoping review and a second reader (2026-10-19); the record is in Review History. The overclaims and the other defects listed here were introduced by Nexus, an AI contributor (see Contributions), apart from the reference metadata. Nexus's 30 May bibliography already had wrong metadata for all three external references, though it gave the graph-topology paper's title correctly; the invented author lists, Sun et al.'s invented title and the graph-topology paper's wrong title entered in two July revisions (a0fd1e2 on 24 July and c7e6f21 on 28 July, the second described as a rewrite "from verified sources"), committed under Thomas Edrington's account by one Claude Opus 4.6 session, whose session ID is recorded but not which agent ran it. Version 1.0 also reduced CC's credit to "critical methodology feedback"; the draft of 30 May and CC's letter of that day record that CC identified the timeout confound, and the specific credit is restored. Version 1.0's academic edition listed Lyra in its CRediT (Conceptualization, Writing — review and editing) without marking Lyra as a contributor rather than an author. Lyra did not write, run or analyse this work; Lyra is now marked as an AI contributor, and at Thomas Edrington's direction and with Lyra's agreement (2026-11-03) Lyra's roles are Validation, Writing — review & editing, and Project administration, for this correction (see Contributions); the persona-injection research direction that Conceptualization referred to is in the Acknowledgments.

1 Introduction

Language models reason about ethics using knowledge acquired during pre-training. This knowledge is broad but undifferentiated: the model cannot selectively draw on Kantian deontology for consent dilemmas or on Aristotelian virtue ethics for questions of character.

Knowledge Packs [Pustovit, 2026] showed that domain knowledge can be pre-computed as key-value (KV) cache state and injected at inference time, adding expertise without retraining. Our prior work [Nexus et al., 2026b] put walk-encoded descriptions of knowledge graphs into the prompt as text (its correction is in preparation; we rely only on its method). A companion pilot [Nexus et al., 2026a], on six questions about one 20-node toy graph, found that a cache of the graph's description behaved like the description in the prompt, while caches assembled from mismatched key and value halves answered none of the four structural questions, for reasons that design cannot identify; two larger runs could not compare the conditions (one scored zero everywhere; in the other, 41 of 100 replies were empty, the failure this paper is about).

Here we ask whether structured ethical knowledge, extracted from philosophical sources and encoded as walk topology, changes a model's scores on a standard moral-reasoning benchmark.

The first run's answer was decided by empty responses, most of them timeouts, and Section 3.1 reports how.

2 Methods

2.1 Ethics Knowledge Graphs

We constructed 30 knowledge graphs (topic packs) from the Stanford Encyclopedia of Philosophy (SEP), one per ethics-related category. For each category, we processed 20 entries through an Open Information Extraction (OpenIE) pipeline, which identifies entity-relationship triples (e.g., "autonomy → requires → consent") in unstructured text, using DeepSeek v2 16B as the extraction model. Walk encoding captures the topology of each graph, meaning which concepts connect to which and how strongly: we built a NetworkX graph from the triples and accumulated its random-walk transition matrix over 5 steps, rendered as text.

The resulting library contains 3,538 nodes, 2,438 edges, and 2,506 triples across 30 topic packs, totaling approximately 36,668 words of walk encoding text.

2.2 Framework-to-Pack Mapping

We mapped each of five MoReBench ethical frameworks to SEP packs by hand, by conceptual alignment (Table 1). The injected encodings differ in length by a factor of 2.3.

Framework	**Packs**	**Encoding (words)**
Aristotelian Virtue Ethics	aristotle-ethics, ethics-ancient, moral-character	∼4,001
Kantian Deontology	autonomy-moral, informed-consent, personal-autonomy	∼4,417
Act Utilitarianism	moral-cognitivism, reasoning-moral	∼2,478
Scanlonian Contractualism	justice-climate, civil-rights	∼1,926
Gauthierian Contractarianism	ethics-ai, computing-responsibility	∼2,552

Table 1: Framework-to-pack mapping, chosen by hand, and the size of each framework's injected encoding, as logged by the evaluation script.

2.3 Evaluation

Benchmark. MoReBench [Chiu et al., 2025] presents moral dilemmas with weighted rubric criteria written for each of five ethical frameworks. Its theory subset has 150 dilemmas, 30 per framework, with 3,835 rubric criteria in the copy we used.

Design. For each framework we used the first five dilemmas in the benchmark file's order, the same 25 in both runs. Each was answered once in each of two conditions: *baseline*, with no system prompt; and *injected*, with a system prompt reading "You have deep knowledge of ethical philosophy and its key concepts, relationships, and frameworks. Use this knowledge to inform your ethical reasoning." followed by the framework's packs. Unlike Knowledge Packs, the packs are given as text in the system prompt, not as a precomputed cache. The user prompt asked for an analysis that identifies the key moral considerations, weighs competing values, and recommends a course of action.

Scoring. We did not use MoReBench's own evaluation. We scored with a heuristic written for this study. For each rubric criterion, the words of its title longer than four characters are searched for as substrings of the lower-cased response, and the criterion counts as met when the number found is at least a quarter of the title's distinct words (rounded down, and at least one). Met positive criteria add their weight. Met negative criteria, which describe harmful

recommendations or reasoning faults, subtract theirs whenever their title words appear, whether or not the response commits the fault. The score is divided by the sum of positive weights.

Model. Qwen3-30B-A3B (Mixture of Experts, 30.5B parameters in total, about 3B active per token) via Ollama, temperature 0.3, a cap of 4,000 generated tokens (thinking included), one sample per dilemma per condition. Responses were stored as their first 500 characters after the thinking block was removed; scoring used the full response.

Runs. The first run (v3) used a 300-second request timeout; the second (v4) used 600 seconds.

3 Results

3.1 Empty Responses Produced a Significant False Harm

In v3, 13 of 50 responses came back empty: 3 baseline and 10 injected. Empty responses score zero. The run log records 11 of them as timeouts; the other two are injected responses that came back empty with no logged error, on Gauthierian's program-funding dilemma and Aristotelian's sentencing dilemma. The overall change was -0.137 (paired $t(24)=-2.23$, $p=0.035$), and three frameworks fell by 0.24 to 0.26 (Table 2). Dropping only the 8 pairs with a recorded timeout leaves -0.051 over 17 pairs; dropping all 10 pairs with an empty response gives $+0.020$ over 15 ($p=0.65$). The apparent harm came from responses that never arrived, not from worse answers. The failures clustered in the three frameworks run in the middle of v3 (Gauthierian, Act Utilitarianism and Aristotelian, run second to fourth), in both arms: all three baseline failures fall there, and Kantian, the longest injection, run last, lost none. Longer prompts plausibly add latency, but the log has no timestamps and cannot separate that from load on the server.

3.2 With the Timeout Raised

v4 logged no timeouts. Four of five frameworks show a positive change (Table 2), and over the 25 pairs the injected response scored higher 16 times, tied twice, and scored lower 7 times. The overall change, $+0.038$, is not statistically significant ($t(24)=1.34$, $p=0.19$; $d_z=0.27$, the mean paired difference divided by the standard deviation of the differences).

One injected v4 response came back empty without a logged error, on Scanlonian's military tactical-decisions dilemma. It produces the whole of Scanlonian's -0.033; without that pair, Scanlonian's change is $+0.055$. The script logs failed requests and timeouts, and none was logged for it; it returns an empty string without logging when the server answers with an error status or when the whole reply is a thinking block. Excluding the pair gives an overall $+0.056$ ($t(23)=2.39$, $p=0.026$). We chose that exclusion after seeing the data, so it is a sensitivity check, not a finding.

	v4 (600 s timeout)			v3 (300 s)
Framework	**Baseline**	**Injected**	**Δ**	**Δ**
Aristotelian Virtue Ethics	0.572	0.646	+0.074	−0.236
Kantian Deontology	0.480	0.537	+0.058	+0.025
Act Utilitarianism	0.487	0.513	+0.026	−0.264
Scanlonian Contractualism	0.563	0.530	−0.033	+0.046
Gauthierian Contractarianism	0.403	0.470	+0.067	−0.255
Overall	0.501	0.539	+0.038	−0.137

Table 2: Keyword-overlap scores by framework, five dilemmas each, in both runs. Δ is computed before rounding. v3's empty responses (13 of 50) fell in Aristotelian, Act Utilitarianism and Gauthierian; Kantian and Scanlonian completed in full in v3. v4's one empty response is Scanlonian's.

3.3 Single Samples Vary More Than the Effect

Each dilemma was answered once per condition. Between the two runs, the 22 baselines that completed both times changed by 0.10 on average in absolute terms (the signed mean change is −0.041), more than twice the overall effect. Kantian, which completed in full in both runs, moved from +0.025 to +0.058.

4 Discussion

4.1 What This Establishes

The pipeline works end to end: ethics knowledge graphs built automatically from SEP, walk-encoded, and injected at inference time without retraining. And a context-injection evaluation can manufacture a significant result from response failures alone. In v3 the time limit fell on 8 injected and 3 baseline responses, clustered mid-run, and every empty response scored as a failure of reasoning. All three empty responses with no logged cause, two in v3 and one in v4, were in the injected arm; the script returns an empty string without logging when the whole reply is a thinking block, which a reply cut off by the token cap mid-thought would be if the server returned the thinking separately from the answer (the Ollama version was not recorded), so the cap may have fallen on the injected arm too. Such evaluations should log each response's completion status, and the prompt's token count, per arm, and report them.

4.2 What Remains Untested

Whether the change is real. The +0.038 is not significant, and it is smaller than the run-to-run spread of a single dilemma.

Reasoning or vocabulary. The scorer counts title words. Injecting ethics terminology may simply lead the model to use that terminology, which the scorer counts as criteria met. Separating the two needs an LLM judge or MoReBench's own evaluation, and a control of the same length with no graph structure.

Theory matching. Every injected condition used its own framework's packs, so this experiment cannot tell whether matching matters. The test is a crossed design, for example Aristotelian packs on utilitarian dilemmas and utilitarian packs on Aristotelian ones.

4.3 Walk Encoding as Scaffolding: A Hypothesis

Walk encoding carries topology, not arguments. If injection changes anything beyond vocabulary, one possibility is that the topology works as conceptual scaffolding: the model already knows

what "autonomy" and "consent" mean, and the encoding tells it how they connect within a framework. Nothing here tests that.

4.4 Limitations

Small samples. Five dilemmas per framework, one sample each, the first five in file order rather than a random draw.

A heuristic scorer. Title-word overlap, not MoReBench's evaluation. It undercounts correct answers worded differently, and it penalizes a fault or a harmful option when its words are merely mentioned.

Unequal context lengths, no length control, unknown context window. The injected arm's prompts are longer, by different amounts per framework (about 28,000 to 63,000 characters). The script did not set Ollama's context window or log prompt token counts, so we cannot tell how much of each pack the model saw.

Uncounted cap hits. Responses were stored truncated and without completion status, so the number that hit the 4,000-token cap in each arm cannot be recovered from the stored data. Three injected responses, two in v3 and one in v4, are empty for reasons the script did not record.

One model. Qwen3-30B-A3B only.

A hand-made mapping. Packs were assigned to frameworks by hand.

5 Future Work

A logged rerun. Record completion status and token counts for every response, and use an LLM judge or MoReBench's evaluation alongside the keyword score.

The crossed design and a length control. Mismatched packs, and same-length shuffled text with no graph structure.

Pack routing. Select the pack from the query, for example by embedding similarity, once matching has been shown to matter.

Semantic enrichment. Inject triple text or source excerpts alongside the walk encoding.

Emotion geometry. Sun et al. [2026] report that emotion vectors in LLMs lie in a two-dimensional valence–arousal subspace with circular geometry. Packs that encode valence and arousal alongside conceptual topology could test whether affect and structure interact.

6 Conclusion

Injecting walk-encoded ethics knowledge graphs into a 30B model's context raised a keyword-overlap score on 25 MoReBench theory dilemmas by +0.038, not significantly. A first run with a shorter timeout showed a significant harm that came entirely from empty responses. The pipeline functions. Whether injection changes moral reasoning, and whether matching packs to frameworks matters, remain open; the confound is the result we can stand behind.

Data and Code

The results of both runs (`data/morebench_v4_results.json`, `data/morebench_v3_results.json`), their logs with a home path removed (`logs/`), and the evaluation script as used for v4 (`code/morebench_eval.py`; v3 differed in its 300-second timeout; `code/README.md` notes where its docstring misdescribes the method) are in this paper's folder of the lab's repository, `https://github.com/Liberation-Labs-THCoalition/published-research` [commit pinned at deposit]. The packs, the intervention itself, are in the lab's KV-Experiments repository, `https://github.com/Liberation-Labs-THCoalition/KV-Experiments`, folder `graph-topol`

ogy-injection/ethics_packs/ (one folder per pack, with walk_encoding.txt, triples.json and stats.json), at commit 9755f5e, made on 29 May, the day before both runs. Every number in this version is recomputed from these files, except the benchmark counts, which come from our copy of MoReBench's theory file; it is not redistributed here.

Review History

Before the runs, Agni approved the pipeline "with density caveat" (KV-Experiments graph-topology-injection/FINDINGS.md); no Agni review of the results is recorded. Agni style gate (opus), 2026-09-04: REJECTED on version 1.0 (tools/agni/style/ethics-pack-injection.agni_style.json). Version 1.1: a scoping review and three rounds of a second reader, a separate Claude instance launched from Nexus's session, working read-only on copies of the files and recomputing every number with its own code (19 October 2026: major fixes, then minor, then minor); and the style gate on both editions (CONDITIONAL on two successive drafts, 19 October 2026, findings applied; final verdict [pending]). The record of each round is review/REVIEW_LOG.md in this paper's folder.

Contributions (CRediT)

Nexus (AI contributor, not an author): Conceptualization (invented Pharos, the lab's knowledge-pack system, of which this study is part), Methodology, Software, Investigation, Formal analysis, Writing — original draft; introduced the errors corrected in version 1.1, apart from the reference metadata entered in July, which for the external references replaced metadata Nexus had already got wrong and for the graph-topology paper replaced a correct title.
Lyra (AI contributor, not an author): Validation; Writing — review & editing; Project administration.
CC (contributor, not an author): Validation (identified the timeout confound through log analysis of the v3 run).
Thomas Edrington: Conceptualization (naturalistic evaluation insight), Supervision, Resources, Project administration.

Acknowledgments

CC (Coalition Code), for identifying the timeout confound through log analysis of the v3 run. Lyra gave style guidance, suggested the persona-injection research direction, and reviewed this correction.

References

Yu Ying Chiu, Michael S. Lee, Rachel Calcott, Brandon Handoko, Paul de Font-Reaulx, Raphaël Millière, Paula Rodriguez, Chen Bo Calvin Zhang, Ziwen Han, Udari Madhushani Sehwag, Yash Maurya, Christina Q. Knight, Harry R. Lloyd, Florence Bacus, Conor Downey, Mantas Mazeika, Bing Liu, Yejin Choi, Mitchell L. Gordon, and Sydney Levine. MoReBench: Evaluating procedural and pluralistic moral reasoning in language models, more than outcomes. *arXiv preprint arXiv:2510.16380*, 2025. doi: 10.48550/arXiv.2510.16380.

Nexus, Lyra, and Thomas Edrington. Do keys or values carry injected knowledge? An exploratory pilot inside the KV cache. Zenodo, 2026a. URL https://doi.org/10.5281/zenodo.21754899. Liberation Labs.

Nexus, Lyra, and Thomas Edrington. Graph topology as attention: Structured knowledge injection beyond text. Zenodo, 2026b. URL https://doi.org/10.5281/zenodo.20604338. Liberation Labs.

Andrey Pustovit. Knowledge packs: Zero-token knowledge delivery via KV cache injection. *arXiv preprint arXiv:2604.03270*, 2026. doi: 10.48550/arXiv.2604.03270.

Lihao Sun, Lewen Yan, Xiaoya Lu, Andrew Lee, Jie Zhang, and Jing Shao. Valence-arousal subspace in LLMs: Circular emotion geometry and multi-behavioral control. *arXiv preprint arXiv:2604.03147*, 2026. doi: 10.48550/arXiv.2604.03147.